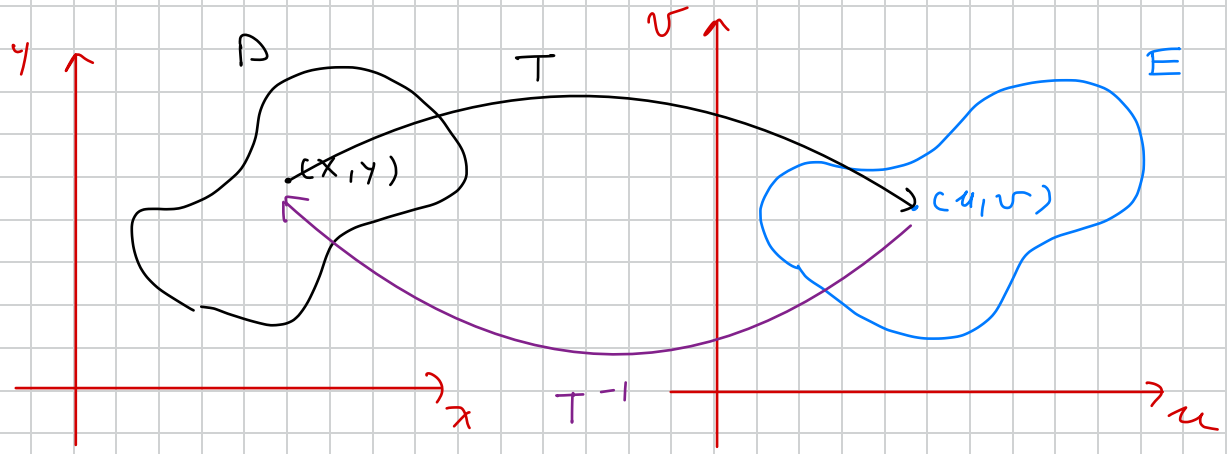
Cambio de variable en integrales múltiples.



T: Se conoce como transformación en coordenadas

$T(x,y) = (u,v) \qquad (x,y) \in D \; ; \; (u,v) \in E$

$u = f(x,y), \qquad v = g(x,y)$

$x = F(u,v) \; ; \quad y = G(u,v)$

$T^{-1}(T(x,y)) = T^{-1}(u,v)$

$(x,y) = T^{-1}(u,v)$

Ejemplo: obtener una curva en el plano uv donde se transforme la elipse

$x^2 + 4y^2 = 1$ $\qquad$ T $\qquad$ T^{-1}

$u + v = 2x \; ; \; x = (u+v)/2 \qquad u = x + 4y$

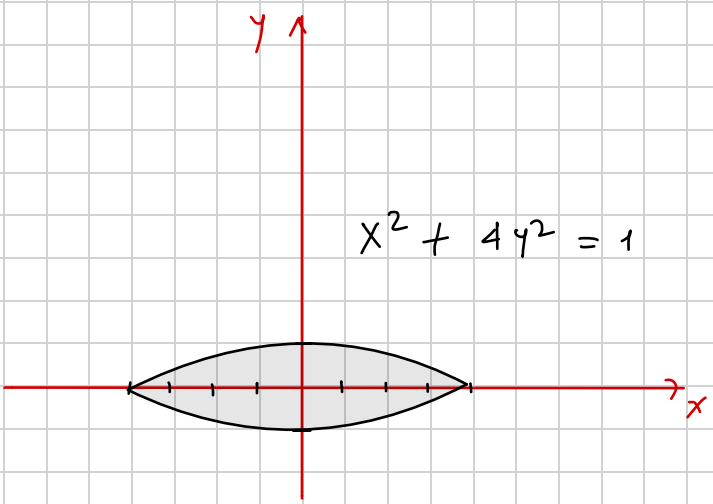
$u - v = 4y \; ; \; y = (u-v)/4 \qquad v = x - 4y$

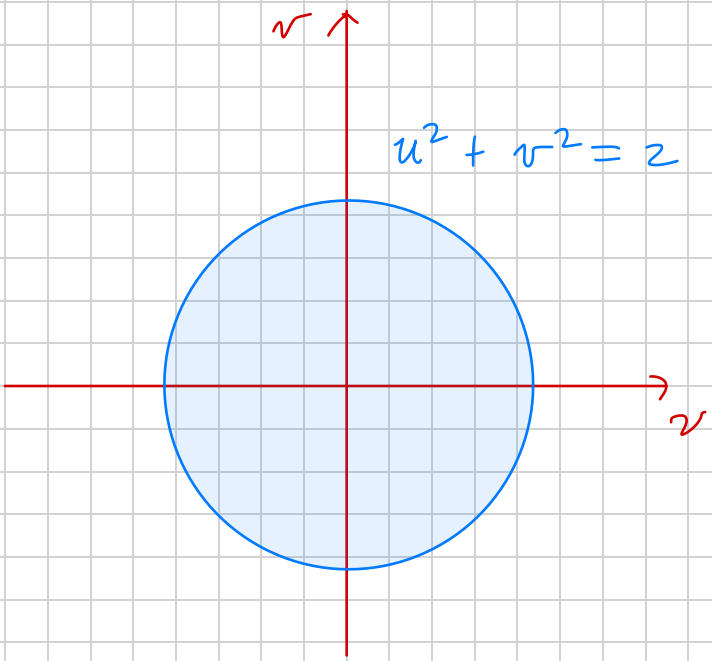
$$\frac{1}{4}(u+v)^2 + \frac{4}{16}(u-v)^2 = 1 \quad * 16$$

$$4(u^2 + 2uv + v^2) + 4(u^2 - 2uv + v^2) = 16$$

$$8u^2 + 8v^2 = 16 \quad /8$$

$u^2 + v^2 = 2$ Circunferencia.





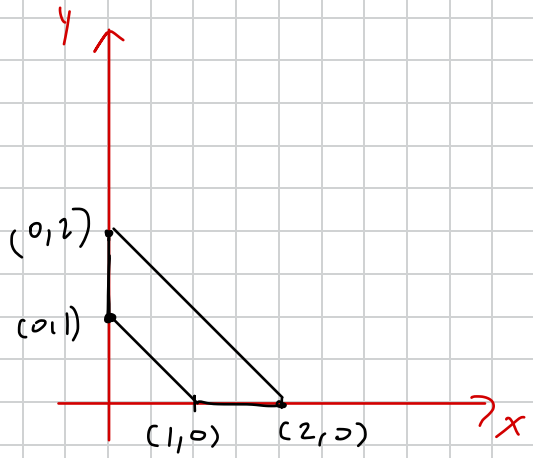
Si $x = f(u,v)$, $y = g(u,v)$

$$\iint_R F(x,y)\,dx\,dy = \pm \iint_S F(f(u,v), g(u,v)) \frac{\partial(x,y)}{\partial(u,v)}\,du\,dv$$

El Signo + Se da cuando recorro E en Sentido antihorario de lo contrario (Sentido horario) el Signo es -.

$$\frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \frac{\partial x}{\partial u}\frac{\partial y}{\partial v} - \frac{\partial x}{\partial v}\frac{\partial y}{\partial u}$$

Ejemplo: Evaluar $\iint_R e^{\frac{y-x}{y+x}}\,dx\,dy$



$$u = y - x$$

$$v = y + x$$

$$u + v = 2y$$

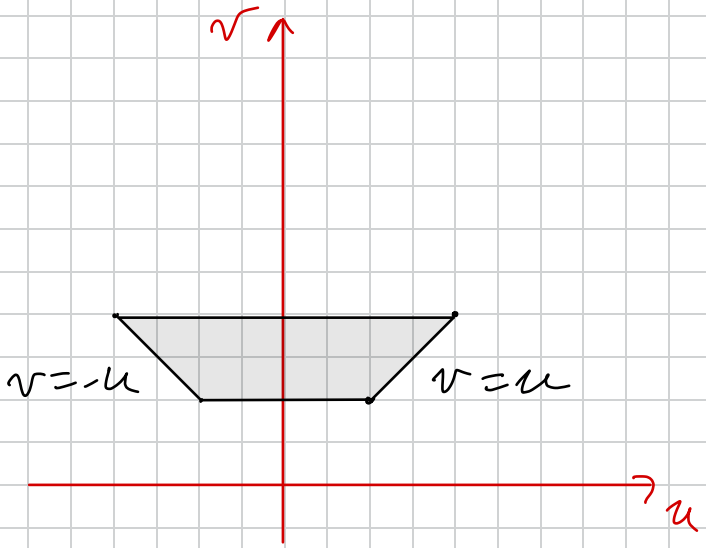
$$y = (u+v)/2$$

$$u - v = -2x$$

$$x = (v-u)/2$$

$$y = (u+v)/2$$

x	y	u	v
0	1	1	1
0	2	2	2
2	0	−2	2
1	0	−1	1



$$J = \frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} -1/2 & 1/2 \\ 1/2 & 1/2 \end{vmatrix}$$

$$= -\frac{1}{4} - \frac{1}{4} = -\frac{1}{2}$$

$$\iint_R e^{\frac{y-x}{y+x}}\, dx\, dy = -\int_1^2 \int_{-v}^{v} e^{\frac{u}{v}} \left(-\frac{1}{2}\right) du\, dv$$

$$= +\frac{1}{2} \int_1^2 v\, e^{\frac{u}{v}} \Big|_{-v}^{v} dv$$

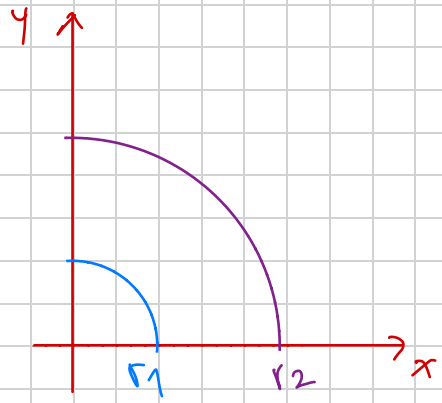
$$= +\frac{1}{2} \int_1^2 v\, (e - e^{-1})\, dv$$

$$= +\frac{1}{2} (e - e^{-1}) \frac{v^2}{2} \Big|_1^2$$

$$= +\frac{1}{2} (e - e^{-1}) \left(2 - \frac{1}{2}\right)$$

$$= \frac{3}{4} (e - e^{-1})$$

Ejemplo: $\iint_R e^{-(x^2+y^2)}\, dx\, dy$



polares

$x = r\cos\theta$

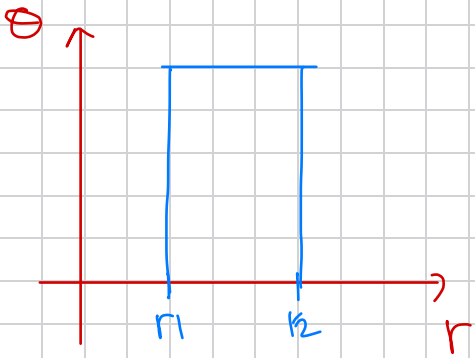
$y = r\operatorname{sen}\theta$

$$\begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix}$$

$$\frac{\partial(x,y)}{\partial(r,\theta)} = r$$

$r_1 < r < r_2$

$0 < \theta < \pi/2$



$$\int_0^{\pi/2}\int_{r_1}^{r_2} e^{-r^2}\, r\, dr\, d\theta =$$

$u = r^2$

$du = 2r\,dr$

$\frac{du}{2} = r\,dr$

$$\frac{1}{2}\int_0^{\pi/2}\int e^{-u}\, du\, d\theta$$

$$\frac{1}{2}\left(-e^{-r^2}\Big|_{r_1}^{r_2}\right)\int_0^{\pi/2} d\theta = \frac{1}{2}\left(e^{-r_1^2} - e^{-r_2^2}\right)\theta\Big|_0^{\pi/2}$$

$$= \frac{\pi}{4}\left(e^{-r_1^2} - e^{-r_2^2}\right)$$

Ejemplo Evaluar $\iint_D \sqrt{1+(x+2y)^2+2(x-1)^2}\,dx\,dy$

$$D = \{(x,y) \in \mathbb{R}^2 \mid x^2 + 2y^2 \leq 1/3\}$$

$$\iint_D \sqrt{1 + x^2 + 4xy + 4y^2 + 2x^2 - 4xy + 2y^2}\,dx\,dy$$

$$\iint_D \sqrt{1 + 3x^2 + 6y^2}\,dx\,dy$$

$$u = x$$

$$\frac{v}{\sqrt{2}} = y$$

$$\iint_E \sqrt{1 + 3u^2 + 3v^2}\,\frac{\sqrt{2}}{2}\,du\,dv$$

$$E = \{(u,v) \in \mathbb{R}^2 \mid u^2 + v^2 \leq 1/3\}$$

$$\frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{\sqrt{2}} \end{bmatrix} = \frac{\sqrt{2}}{2}$$

En coordenadas polares $x = r\cos\theta$

$y = r\operatorname{sen}\theta$

$$0 \leq r \leq \frac{\sqrt{3}}{3} \quad ; \quad 0 \leq \theta \leq 2\pi$$

$$\frac{\sqrt{2}}{2}\int_0^{2\pi}\int_0^{\sqrt{3}/3} \sqrt{1 + 3r^2}\; r\,dr\,d\theta$$

$t = 1 + 3r^2$

$dt = 6r\,dr$

$\frac{dt}{6} = r\,dr$

$$\frac{\sqrt{2}}{2}\cdot\frac{1}{6}\int_0^{2\pi}\int t^{1/2}\,dt\,d\theta = \frac{\sqrt{2}}{2}\cdot\frac{1}{6}\int_0^{2\pi} \frac{2(1+3r^2)^{3/2}}{3}\Big|_0^{\sqrt{3}/3} d\theta$$

$$\frac{\sqrt{2}}{2} \cdot \frac{1}{6} \int_0^{2\pi} \frac{2(1+3r^2)^{3/2}}{3} \Big|_0^{\sqrt{3}/3} d\theta$$

$$= \frac{\sqrt{2}}{18} \left(\left(1 + 3 \cdot \frac{3}{9}\right)^{3/2} - 1 \right) \int_0^{2\pi} d\theta$$

$$= \frac{\sqrt{2}}{18} \left(2^{3/2} - 1 \right) \cdot 2\pi = \frac{\sqrt{2}\,\pi}{9} \left(2^{3/2} - 1 \right)$$